ACADEMICS · UG CURRICULUM

B.Tech EE — IC Design & Technology

4-year interdisciplinary programme · 2025 curriculum onwards

← All UG Programmes
Semester 1
Semester 2
Semester 3
Semester 4
Semester 5
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Semester 7
Semester 8
Basic Science Basket

This curriculum applies to students admitted from 2025–26 onwards. If you joined between 2022 and 2024, refer to the 2022–2024 curriculum page.

SEMESTER 1 (AUGUST – DECEMBER)

CODE	COURSE NAME	CREDITS	TYPE
BT1010	Introduction to Life Sciences	1	Basic Sciences
EE1101	Circuits and Network Analysis	3	Dept. Core
EE2100	Matrix Theory	3	Basic Sciences
ID1063	Introduction to Programming	3	Basic Engg. Skills
LA1760	Communication Skills	2	Liberal Arts
ID1006	Ethics for Engineering Education	1	Liberal Arts
EE1010	Mathematics for Electrical Engineers	3	Basic Sciences
Total		**16**	

SEMESTER 2 (JANUARY – MAY)

CODE	COURSE NAME	CREDITS	TYPE
EE1200	Electrical Circuits Lab	2	Dept. Core
EE1201	Digital Systems	3	Dept. Core
EE1500	Introduction to IC Design & Technology	1	Dept. Core
EE1204	Engineering Electromagnetics	2	Dept. Core
EE1501	Digital Systems Lab	1	Dept. Core
EE1080	Probability	3	Basic Engg. Skills
EE1060	Differential Equations and Transforming Techniques	3	Basic Sciences
ID1054	Digital Fabrication	2	Basic Engg. Skills
Total		**17**	

SEMESTER 3 (AUGUST – DECEMBER)

CODE	COURSE NAME	CREDITS	TYPE
EE2000	Signal Processing	3	Dept. Core
EE2104	Semiconductor Device Fundamentals	3	Dept. Core
EE2501	Semiconductor Devices Lab	2	Dept. Core
EE2503	Numerical Methods for Device Modeling	2	Dept. Core
EP/PH/CY xxxx	*Basic Science Basket (4 credits)*	*4*	*Basic Sciences*
ID2230	Data Structures & Applications	3	Basic Engg. Skills
EE2510	Productivity Tools for IC Design	1	Dept. Core
Total		**18**	

SEMESTER 4 (JANUARY – MAY)

CODE	COURSE NAME	CREDITS	TYPE
EE2400	Analog Electronics	3	Dept. Core
EE2401	Analog Lab	2	Dept. Core
EE2504	Electronic System Design Project Lab	2	Dept. Core
EE2520	Introduction to CMOS Processing	3	Dept. Core
EE2530	Microprocessor and Computer Architecture	3	Dept. Core
LA xxxx	*Liberal Arts Elective*	*1*	*Liberal Arts*
EE2500	Control Theory for Circuits	2	Dept. Core
MA xxxx	*Basic Science Basket*	*1*	*Basic Sciences*
Total		**17**	

SEMESTER 5 (AUGUST – DECEMBER)

CODE	COURSE NAME	CREDITS	TYPE
EE3140	Waveguides and Antennas	3	Dept. Core
EE3150	CMOS Digital Design	3	Dept. Core
EE3502	IC Design Tapeout	2	Dept. Elective
EE3510	RTL Design and Verification	2	Dept. Elective
EE3520	Analog IC Design	3	Dept. Core
EE2080	Random Processes	1	Basic Engg. Skills
EE4500	Embedded Systems	3	Dept. Core
Total		**17**	

SEMESTER 6 (JANUARY – MAY)

CODE	COURSE NAME	CREDITS	TYPE
EE4xxx	Mesoscopic Device Electronics	3	Dept. Core
EE xxxx	*Departmental Elective*	*3*	*Dept. Elective*
EE2802	Fundamentals of Machine Learning	3	Basic Engg. Skills
EE4502	IC Design Test and Characterisation	2	Dept. Core
EE4501	Fabrication & Characterization Lab	2	Dept. Core
LA xxxx	*Liberal Arts Elective*	*2*	*Liberal Arts*
Total		**15**	

SEMESTER 7 (AUGUST – DECEMBER)

CODE	COURSE NAME	CREDITS	TYPE
EE xxxx	*Departmental Elective (or Internship equivalent)*	*6*	*Dept. Elective*
XX xxxx	*Free Elective*	*6*	*Free Elective*
LA xxxx	*Liberal Arts Elective*	*2*	*Liberal Arts*
Total		**14**	

SEMESTER 8 (JANUARY – MAY)

CODE	COURSE NAME	CREDITS	TYPE
LA xxxx	*Liberal Arts Elective*	*2*	*Liberal Arts*
XX xxxx	*Free Elective*	*6*	*Free Elective*
EE xxxx	*Departmental Elective*	*6*	*Dept. Elective*
Total		**14**	

BASIC SCIENCE BASKET (4 CREDITS IN SEMESTER 3 — CHOOSE FROM APPROVED LIST)

CODE	COURSE NAME	CREDITS
MA2140	Introduction to Statistics	1
BM1000	Physio-anatomy I	2
BM1010	Physio-anatomy II	2
BM2053	Mathematical Models & Systems Biology	2
BT1010	Introduction to Life Sciences	1
BT1020	Basic Bioinformatics	2
BT1110	Introduction to Bio-nanotechnology	1
IC1010	Industrial Organic Chemistry I	3
CY1010	Environmental Chemistry	2
CY1031	Chemistry Lab	2
CY1018	Environmental Chemistry (alt.)	2
EP1108	Modern Physics	2
EP1118	Maths for Physics	2
EP1031	EP Lab I	2
EP/PH/CY/MA	Any approved course with non-overlapping content	